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Hydrophilic Urinary Catheter Products with Microcapsules of Anti-bacterial Agents

Abstract

A catheter product including a urinary catheter with anti-microbial microcapsules. More particularly a urinary catheter with anti-microbial microcapsules that release the anti-microbial agent upon exposure to irradiation.

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Background/Summary

[0001] The present application claims the benefit of and priority to U.S. Provisional Application No. 63/339,172, filed May 6, 2022, which is hereby incorporated herein by reference.

FIELD OF THE INVENTION

[0002] This disclosure generally relates to a urinary catheter product including anti-microbial microcapsules, wherein the microcapsules are incorporated in a hydrophilic coating or in a hydration solution of the product. More particularly, this disclosure relates to urinary catheter products with microcapsules that release an anti-microbial agent when exposed to irradiation.

BACKGROUND

[0003] Intermittent catheterization is a good option for many users who suffer from various abnormalities of the urinary system. With the advent of intermittent urinary catheters, individuals with problems associated with the urinary system can conveniently self-catheterize to drain the individual's bladder. Individuals who suffer from urinary incontinence will self-catheterize several times a day.

[0004] Although catheters are typically prepared in sterile environments and are provided with safe handling instructions, catheter users are at risk of contracting a urinary tract infection due to several different factors. For example, the catheters may become contaminated during use and introduce bacteria into the urethra. The catheter also may carry bacteria along the urinary tract.

[0005] Therefore, there is a need for methods and devices that allow catheter users to prevent urinary tract infections.

SUMMARY OF INVENTION

[0006] There are several aspects of the present subject matter which may be embodied separately or together in the subject matter claimed below. These aspects may be employed alone or in combination with other aspects of the subject matter described herein, and the description of these aspects together is not intended to preclude the use of these aspects separately or the claiming of such aspects separately or in different combination as set forth in the claims appended hereto.

[0007] In one aspect, a urinary catheter product is provided. The urinary catheter product includes a package containing a urinary catheter and a hydration medium. The urinary catheter includes a catheter tube having a proximal insertion end and a distal drainage end. The urinary catheter further includes a hydrophilic polymer coating on an outer surface of the catheter tube. The urinary catheter product also includes a plurality of microcapsules inside the package, wherein the microcapsules include a polysaccharide encapsulating an anti-microbial agent. The microcapsules are configured to release the anti-microbial agent upon irradiation.

[0008] In another aspect, a urinary catheter product is provided. The urinary catheter product includes a package containing a urinary catheter and a hydration medium. The urinary catheter includes a catheter tube having a proximal insertion end and a distal drainage end. The urinary catheter further includes a hydrophilic polymer coating on an outer surface of the catheter tube. The urinary catheter product also includes a plurality of microcapsules inside the package, wherein the microcapsules include an encapsulate encapsulating an anti-microbial agent. The microcapsules are configured to release the anti-microbial agent upon exposure to a pH range.

[0009] In another aspect, a urinary catheter product is provided. The urinary catheter product includes a package containing a urinary catheter and a hydration medium. The urinary catheter includes a catheter tube having a proximal insertion end and a distal drainage end. The urinary catheter further includes a hydrophilic polymer coating on an outer surface of the catheter tube. The urinary catheter product also includes a plurality of microcapsules inside the package, wherein the microcapsules include an encapsulate encapsulating an anti-microbial agent. The microcapsules are incorporated into the hydration medium.

[0010] In yet another aspect, a method for making an anti-microbial catheter is provided. The method includes dissolving a polysaccharide into an aqueous solution and adding an anti-microbial agent to the aqueous solution. The method also includes drying the aqueous solution to create a plurality of microcapsules, wherein the microcapsules include the polysaccharide encapsulating the antimicrobial agent. Furthermore, the method includes adding the plurality of microcapsules to a package configured to store the catheter and the hydration medium and releasing the anti-microbial agent.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a plan view of an embodiment of the urinary catheter product;
[0012] FIG. 2 is an enlarged cross-sectional schematic view of an embodiment of the microcapsule;
[0013] FIG. 3 is a side view of a portion of the catheter and an enlarged inset of the catheter outer surface of the urinary catheter product of FIG. 1;
[0014] FIG. 4 is a plan view of another embodiment of the urinary catheter product;
[0015] FIG. 5 is a cross-sectional view of the urinary catheter product of FIG. 4; and
[0016] FIG. 6 is a flowchart setting forth the steps in the method of creating an anti-microbial catheter according to an aspect of the present disclosure.

DETAILED DESCRIPTION

[0017] A more detailed description of the device in accordance with the present disclosure is set forth below. It should be understood that the description of the specific devices below is intended to be exemplary, and not exhaustive of all possible variations or applications. Thus, the scope of the disclosure is not intended to be limiting and should be understood to encompass variations or embodiments that would occur to persons of ordinary skill.

[0018] FIGS. 1 and 4 show a urinary catheter product **10**. The catheter product **10** includes a package **12** containing a urinary catheter **14** and a hydration medium **16**. The package includes a cavity configured to receive the urinary catheter **14** and the hydration medium **16**. In one embodiment the package **12** may be made of first and second opposing sheets **13a** and **13b** (shown in FIG. 5). The sheets may be sealed along the edges, to define the cavity. In some embodiments, the package may have a gripping member **9** to assist the user in holding the package **12**. In some embodiments, the package **12** may have a tear strip (not shown) to assist the user in opening the package **12**. The package may have a generally rectangular shape. Other packages known in the art configured to contain a catheter and hydration medium may be used without departing from the scope of the disclosure.

[0019] FIG. 1 further shows urinary catheter **14** inside the package **12**. Urinary catheter **14** includes a catheter tube **18** having a proximal insertion end **20** and a distal drainage end **22**. A lumen extends from the proximal insertion end **20** to the distal drainage end **22**. The proximal insertion end **20** includes one or more openings **15** for receiving urine in communication with the lumen. In one embodiment, the one or more openings **15** for receiving urine may be an eyelet. In another embodiment, the one or more openings **15** for receiving urine may be multiple eyelets. The one or more openings **15** for receiving urine may vary without departing from the scope of the disclosure.
[0020] Furthermore, the distal drainage end **22** of the catheter tube **18** includes a drainage opening for draining urine from the lumen. In one embodiment, the distal drainage end **22** includes a drainage member **17**. The drainage member **17** may be, for example, but not limited to, a funnel or connector for connection to a drainage bag (not shown). Other drainage members known in the art may be used without departing from the scope of the disclosure.

[0021] The urinary catheter tube **18** includes a hydrophilic polymer coating **24** (FIGS. 3 and 5) on an outer surface **26** of the catheter tube **18**. The hydrophilic polymer coating **24** is configured to

absorb the hydration medium **16** to lubricate the catheter to aid in insertion into a urethra. Before packaging, the catheter tube **18** may be coated with a hydrophilic polymer solution and then cured to form the hydrophilic coating **24**.

[0022] The hydrophilic coating solution may include a suitable solvent (such as water and/or ethanol), hydrophilic polymer(s) and optional additives. The resulting hydrophilic coating may include any suitable hydrophilic polymer, such as polyvinylpyrrolidone, polyethylene oxide or polyvinyl alcohol. Examples of hydrophilic polymer solutions used to create a hydrophilic polymer coating and hydrophilic coatings are disclosed in WO2016/168461, filed Apr. 14, 2016, to Hollister. Such hydrophilic coating solutions may include a solvent, hydrophilic polymer(s), other monomers/oligomers and optional additives. In one alternative, the hydrophilic coatings may include a hydrophilic polymer and polyethylene glycol diacrylate having a number average molecular weight of less than about 1000. In one embodiment the number average molecular weight may be less than about 600. In one alternative, the hydrophilic coating comprises about 80 wt % to about 99.5 wt % hydrophilic polymer and about 0.5 wt % to about 20 wt % polyethylene glycol diacrylate. Other hydrophilic polymer solutions and coatings may be used without departing from the scope of the disclosure. During the curing process, the hydrophilic polymer solution crosslinks to create a hydrophilic polymer coating **24** containing a polymer matrix **25** with interstices **27** (shown in FIG. 3).

[0023] A plurality of microcapsules **28** is located in the package **12** of the urinary catheter product **10**. In one embodiment, the microcapsules **28** are incorporated in the hydrophilic polymer coating **24**. In another embodiment the microcapsules **28** are incorporated in the hydration medium **16**. In yet, another embodiment, the microcapsules are incorporated into both the hydrophilic polymer coating **24** and the hydration medium **16**. Additionally, the hydration medium may be a carrier that carries the microcapsules into the matrix of the hydrophilic polymer coating **24**.

[0024] FIG. 2 is a schematic illustration of a microcapsule **28**. The microcapsule **28** includes an encapsulate **30** encapsulating an anti-microbial agent **32**. Preferably, the anti-microbial agent **32** aids in the reduction of urinary tract infections. Upon exposure to a selected condition, the encapsulate **30** breaks down to release the anti-microbial agent **32**. In one alternative, the encapsulate **30** breaks down when exposed to irradiation at a dose between about 25 kGy and about 65 kGy. For example, the encapsulate/microcapsule may liquefy when exposed at a selected radiation dose, thereby releasing the anti-microbial agent. The selected dose may be the same dose used for curing the hydrophilic coating or radiation sterilization of the catheter product. Furthermore, the radiation may be E-beam or gamma. In another alternative, the encapsulate/microcapsule breaks down when exposed to a selected pH. For example, the encapsulate/microcapsule may break down when it is exposed to a pH of about 3.5-7.

[0025] In one embodiment, the anti-microbial agent **32** is a biocide. Suitable biocides may be silver, antibiotics with biocidal properties such as vancomycin, sparfloxacin, minocycline, and rifampin among others, and poly (catechins)-antibiotic conjugates such as polycatechin with trimethoprim (TMP) and/or polycatechin with TMP and sulfamethoxazole (SMZ). In another embodiment, the anti-microbial agent **32** may have steric repulsion, electrostatic repulsion, and/or low surface energy properties. In yet another embodiment, the anti-microbial agent **32** may be an antifouling material. Suitable antifouling materials include, but are not limited to, hydrophilic materials such as SAM-OEG, PEG, POEGMA; polyzwitterions such as polyMPC, polyCBMA, polySBMA; phosphorylcholines; sulfobetaines; and carboxybetaines. In yet another embodiment, the anti-microbial agent **32** may be any other suitable anti-microbial agent, such as chlorhexidine, amphotercin, bacteriophages, nitrofur, triclosan, anti-microbial peptides, and/or allantoin. In another embodiment, the anti-microbial agent **32** may be an enzyme. Suitable enzymes include hydrolytic enzymes such as proteolytic enzymes (e.g. subtilisin and lysostaphin), polysaccharide hydrolyzing enzymes (e.g. alpha amylase, dispersin B, chitinases, beta-glucanases, lysozyme, and alginate lyase), DNases, and bacteriophage lysins. In yet another embodiment, the anti-microbial

agent **32** may be nitric oxide. In yet another embodiment, the anti-microbial agent **32** may be liposomes such as spherical lipid vesicles that consist of one or more phospholipid bilayers. Other anti-microbial agents known in the art may be used without departing from the scope of the disclosure.

[0026] The encapsulate **30** may be made of a variety of materials, such as starch, starch derivatives, proteins, gelatins, gums, lipids, natural and synthetic polymers, or any combination of such encapsulates. Examples of natural polymers include, but are not limited to, protein-based polymers and polysaccharide-based polymers. Suitable protein-based polymers include, but are not limited to, albumin and gelatin. Suitable polysaccharide-based polymers include, but are not limited to, agarose, alginate, hyaluronic acid, dextran, gellan gum, chitosan, among others. Examples of synthetic polymers include, but are not limited to, poly(ϵ -caprolactone), poly(lactic-co-glycolic acid), and polyethylene glycol. Other encapsulates known in the art may be used without departing from the scope of the disclosure.

[0027] In one alternative, the microcapsules **28** can be made by dissolving the polysaccharide in a solute to form a homogenous solution. In one alternative, the polysaccharide in an amount greater than about 0.5% w/w or about 1.0% w/w, for example, between about 0.5% w/w and about 3% w/w may be dissolved in water. The solution may be heated to assist in dissolving the polysaccharide in the water. For example, gellan gum in an amount greater than about 0.5% w/w or about 1.0% w/w, for example, between about 0.5% w/w and about 3% w/w is dissolved in water at a temperature that is greater than 75° C. Once dissolved, a selected amount/concentration of the anti-microbial agent **32** may be added to the polysaccharide aqueous solution. In one alternative, the selected amount of anti-microbial added to the solution may be between about 1% w/w and about 50% w/w. For example, between about 5% w/w and about 25% w/w of anti-microbial may be added to a gellan gum solution. After the anti-microbial agent **32** has been added, the solution is dried to encapsulate the anti-microbial agent **32** in the polysaccharide. For example, the solution may be cooled to a selected temperature to solidify the polysaccharide and to form a microcapsule encapsulating the anti-microbial therein. When gellan gum is the polysaccharide, the solution may be cooled to below 37° C., thereby commencing solidification of the gellan gum. To assist with drying the polysaccharide encapsulate at temperatures, spray-drying and spin-drying methods may be employed. Other methods of encapsulation may also be used, such as freeze drying, vacuum drying, extrusion, spray cooling, fluidized bed coating, centrifugal extrusion, co-crystallization, molecular inclusion, pan coating, air suspension coating, coacervation, solvent evaporation, and polymerization among other methods known in the art.

[0028] The microcapsules **28** may be 10 μ m-5 mm. Preferably, the microcapsules **28** are 50 μ m-1 mm. In one embodiment, the size of the microcapsule **28** may be varied based on the pressure of spray-drying the encapsulate **30**. In another embodiment the size of the microcapsule **28** may be varied based on the speed of spin-drying the encapsulate **30**. In a further embodiment, the size of the microcapsule **28** may be reduced by grinding. The smaller size of the microcapsule **28** aids with incorporation into the hydrophilic polymer coating **24** or the hydration medium **16**. The shape of the microcapsule may be spherical or elliptical.

[0029] In one embodiment, the microcapsules **28** may be incorporated into the hydrophilic polymer coating **24** on the outer surface **26** of the catheter tube **18**. The microcapsules **28** may be incorporated into the coating **24** by adding them to a hydrophilic polymer coating solution. The microcapsules may be added to the hydrophilic polymer coating solution at a concentration between 0.01% w/w to 5.0% w/w. The coating solution may include water and/or ethanol along with the hydrophilic polymer and optional additives. When gellan gum is used, at ambient temperature, the gellan gum microcapsules **28** are insoluble in ethanol and mostly insoluble in water and will suspend in the polymeric coating solution primarily due to their outer surface becoming softened and sticky by aqueous fraction.

[0030] After the microcapsules **28** have been suspended in the hydrophilic polymer coating

solution, the outer surface **26** of the catheter tube **18** may be coated. The catheter tube may be coated by dip-coating or spray-coating. Other methods of coating a catheter known in the art may be used without departing from the scope of the disclosure.

[0031] Once the catheter tube **18** has been coated, the catheter **14** may be cured and dried. In an embodiment, the catheter **14** is cured using UV irradiation and dried at 70° C. Other methods of curing and drying may be used without departing from the scope of the disclosure.

[0032] FIG. **3** shows a portion of the catheter tube **18** with an enlarged inset showing the hydrophilic polymer coating **24** with microcapsules **28** incorporated into the coating **24**. During curing, the hydrophilic polymer solution crosslinks **29** to create a hydrophilic polymer coating **24** with a polymer matrix **25** and interstices **27**. Without being chemically bonded to the coating, the microcapsules **28** may be located in the interstices **27** of the polymer matrix **25**.

[0033] In another embodiment, the microcapsules **28** may be incorporated into the hydration medium **16**. FIGS. **4** and **5** show a urinary catheter product **10** including a package **12** containing a urinary catheter **14** and a hydration medium **16**. The hydration medium may be water or an aqueous solution. Other hydration mediums known in the art may be used. In some embodiments the hydration medium **16** includes a liquid. In other embodiments, the hydration medium **16** may include a sticky xanthan gum matrix. Microcapsules **28** may be incorporated into the hydration medium **16** by suspending them in the medium **16**. Preferably, smaller microcapsules **28** are used to aid with suspension of the microcapsules **28** in the hydration medium **16**. Depending on the antimicrobial agent, the microcapsule **28** may be any suitable size. In one alternative, the microcapsules **28** may be less than 500 um. Similarly, the concentration of microcapsules may vary depending on the antimicrobial agent. In one alternative, the microcapsules **28** may be in a concentration range of 0.01% w/w to 10% w/w in the hydration medium **16**.

[0034] A method for making an anti-microbial catheter **14** is further shown in FIG. **6** in steps **34-42**. At step **34**, a polysaccharide is dissolved into an aqueous solution. In one embodiment, the polysaccharide may be gellan gum, which readily dissolves into an aqueous solution above 75° C. Other polysaccharides may be used without departing from the scope of the disclosure.

[0035] After the polysaccharide is dissolved, at step **36** an anti-microbial agent **32** may be added to the aqueous solution. The anti-microbial agent **32** may be any of the anti-microbial agents described herein.

[0036] At step **38**, the aqueous solution with the anti-microbial agent **32** is dried to create a plurality of microcapsules **28**. The microcapsules **28** include the polysaccharide encapsulating the anti-microbial agent **32**. In one embodiment, the solution may be dried by spray-drying or spin-drying. The size of the microcapsules may be varied by adjusting the pressure of spray-drying or the speed of spin-drying. Other drying methods may be used, such as, but not limited to, freeze drying, vacuum drying, extrusion, spray cooling, fluidized bed coating, centrifugal extrusion, co-crystallization, molecular inclusion, pan coating, air suspension coating, coacervation, solvent evaporation, and polymerization among other methods known in the art.

[0037] After the microcapsules **28** are created, at step **40** the microcapsules **28** are added to the package **12** configured to store the catheter **14** and hydration medium **16**. In one embodiment, the microcapsules **28** are added to the package **12** by being incorporated into the hydrophilic polymer coating **24** on the outer surface **26** of the catheter tube **18**. In another embodiment, the microcapsules **28** are added to the package **12** by being incorporated into the hydration medium **16**.

[0038] After the microcapsules **28** have been added to the package **12**, the catheter **14** and hydration medium **16** are sealed in the package **12** and at step **42** the anti-microbial agent **32** may be released from the microcapsule **28**.

[0039] In one embodiment, the anti-microbial agent **32** may be released upon sterilization of the catheter product **10** by irradiation. The catheter may be sterilized using electron-beam irradiation. Upon electron-beam irradiation, the polysaccharide encapsulate **30**, depending on its concentration, will soften or convert to liquid, releasing the anti-microbial agent **32**. In the instance where the

microcapsules **28** are incorporated into the hydrophilic polymer coating **24**, the anti-microbial agent **32** will be released directly onto the outer surface **26** of the catheter tube **18**. In the instance where the microcapsules **28** are incorporated into the hydration medium **16**, the anti-microbial agent **32** will be released into the hydration medium **16** and then absorbed by the outer surface of the hydrophilic polymer coating **24**.

[0040] In another embodiment, the anti-microbial agent **32** may be released when the microcapsule is exposed to an environment with a pH within a range of 3.5-7. Preferably, the anti-microbial agent **32** is released when the microcapsule is exposed to an environment with a pH of 3.9-5. More preferably, the anti-microbial agent **32** is released when the microcapsule is exposed to an environment with a pH of 3.9-4.5. The hydration liquid **16** may have a pH configured to release the anti-microbial agent **32**. In some instances, the pH of the hydration medium may change upon irradiation, resulting in release of the anti-microbial agent **32**.

[0041] In yet another embodiment, the anti-microbial agent **32** may be released using temperature or biodegradation with time.

[0042] It will be understood that the embodiments and examples described above are illustrative of some of the applications of the principles of the present subject matter. Numerous modifications may be made by those skilled in the art without departing from the spirit and scope of the claimed subject matter, including those combinations of features that are individually disclosed or claimed herein. For these reasons, the scope hereof is not limited to the above description but is as set forth in the following claims, and it is understood that claims may be directed to the features hereof, including as combinations of features that are individually disclosed or claimed herein.

Claims

1. A urinary catheter product comprising: a package containing a urinary catheter and a hydration medium; the urinary catheter comprising a catheter tube having a proximal insertion end and a distal drainage end; a hydrophilic polymer coating on an outer surface of the catheter tube; and a plurality of microcapsules inside the package, wherein the microcapsules comprise a polysaccharide encapsulating an anti-microbial agent and the microcapsules are configured to release the anti-microbial agent upon irradiation.
2. The urinary catheter product of claim 1, wherein the plurality of microcapsules is incorporated into the hydrophilic polymer coating.
3. The urinary catheter product of claim 2, wherein the plurality of microcapsules incorporated in the hydrophilic polymer coating is in a concentration ranging from 0.1% w/w to 5.0% w/w.
4. The urinary catheter product of claim 1, wherein the microcapsules are located within a matrix of the hydrophilic polymer coating.
5. The urinary catheter product of claim 4, wherein the microcapsules are located in interstices of the matrix of the hydrophilic polymer coating.
6. The urinary catheter product of claim 1, wherein the plurality of microcapsules is incorporated into the hydration medium.
7. The urinary catheter product of claim 6, wherein the hydration medium includes a liquid and the plurality of microcapsules is suspended in the liquid.
8. The urinary catheter product of claim 6, wherein the plurality of microcapsules is in a concentration ranging from 0.1% w/w to 10.0% w/w.
9. The urinary catheter product of claim 1, wherein the polysaccharide encapsulate comprises gellan gum.
10. The urinary catheter product of claim 1, wherein the agent comprises a biocide, a repulsion agent, an antifouling agent, an anti-microbial, an enzyme, a nitric oxide agent, a liposome, or any combination of such agents.
11. The urinary catheter product of claim 1, wherein the anti-microbial agent is released upon the

microcapsules being exposed to electron-beam irradiation.

12. The urinary catheter product of claim 1, wherein the microcapsules are between 10 μm to 5 mm.

13. The urinary catheter product of claim 1, wherein the microcapsules are 50 μm to 1 mm.

14. A urinary catheter product comprising: a package containing a urinary catheter and a hydration medium; the urinary catheter comprising a catheter tube having a proximal insertion end and a distal drainage end; a hydrophilic polymer coating on an outer surface of the catheter tube; and a plurality of microcapsules inside the package, wherein the microcapsules comprise an encapsulate encapsulating an anti-microbial agent and the microcapsules are configured to release the anti-microbial agent upon exposure to a pH range.

15. The urinary catheter product of claim 14, wherein the encapsulate comprises starch, starch derivatives, proteins, gelatins, gums, lipids, natural polymers, synthetic polymers, or any combination of such encapsulates.

16. The urinary catheter product of claim 14, wherein the encapsulate comprises a polysaccharide.

17. The urinary catheter product of claim 14, wherein the encapsulate is gellan gum.

18. The urinary catheter product of claim 14, wherein the anti-microbial agent is a biocide, a repulsion agent, an antifouling agent, an anti-microbial, an enzyme, a nitric oxide agent, a liposome, or any combination of such agents.

19. The urinary catheter product of claim 14, wherein the plurality of microcapsules is incorporated into the hydrophilic polymer coating.

20.-30. (canceled)

31. A urinary catheter product comprising: a package configured to store a urinary catheter and a hydration medium; the urinary catheter comprising a catheter tube having a proximal insertion end and a distal drainage end; a hydrophilic polymer coating on an outer surface of the catheter tube; and a plurality of microcapsules comprising an encapsulate encapsulating an anti-microbial agent, wherein the microcapsules are incorporated in the hydration medium.

32.-58. (canceled)
